

Towards Free Control Functors

Atlantic Category Theory Seminar

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- 3 Free Precontrol Functors from Monoidal Generators
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Background Material: Free Categories

Underlying Graph. Every (small) category $\mathcal{C}$ has an underlying directed multigraph $U_G(\mathcal{C})$ with vertices $\text{Ob}(\mathcal{C})$ and edges $\text{Mor}(\mathcal{C})$. If $f \in \mathcal{C}(X, Y)$, then f is an edge from X to Y .

Underlying Graph Homomorphism. If $F : \mathcal{C} \rightarrow \mathcal{G}$ is a functor, then $(\text{Ob}(F), \text{Mor}(F))$ is a directed multigraph morphism by functoriality. Then $U_G : \mathbf{Cat} \rightarrow \mathbf{Digraph}$ is functorial.

Free Categories. The functor U_G is the right adjoint to the functor which maps each graph G to the path category on G .

$$\begin{array}{ccc} & F_G & \\ \text{Digraph} & \xrightarrow{\quad} & \text{Cat} \\ & U_G & \end{array}$$

Background Material: Monoid Presentations

Free Monoids. Given a set X , the free monoid generated by X is the set X^* of sequences over X with concatenation.

Quotient Monoids. Let M be a monoid and R a submonoid of $M \times M$. Then M/R defined is as following coequalizer.

$$R \xrightarrow{i} M \times M \xrightleftharpoons[\pi_2]{\pi_1} M \xrightarrow{\pi_R} M/R$$

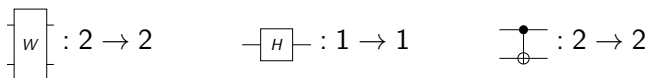
Monoid Presentations. A presentation of a monoid M is a set X and a set $E \subseteq X^* \times X^*$ such that $M \cong X^*/E^*$, denoted $\langle X \mid E \rangle$.

Congruence Relations. A congruence relation ($\approx$) on a monoid M is an equivalence relation on M such that for each $x \in M$ and $y \in M$, $u \approx v$ implies $xuy \approx xvy$. An important example:

$$(\approx_E) = \{(u, v) \in M \times M : \pi_E(u) = \pi_E(v)\}$$

Circuits and Monoidal Categories

Circuits are made up of gates with inputs and outputs.



The inputs and outputs of gates may be typed/sorted.

$$\text{---} \boxed{\text{---}} \text{---} : \mathbf{QWire} \rightarrow \mathbf{CWire}$$

Gates can be composed sequentially and in parallel, with both compositions being associative and unital.

$$y \text{---} \boxed{D} \text{---} z \circ x \text{---} \boxed{C} \text{---} y := x \text{---} \boxed{C} \text{---} \boxed{D} \text{---} z \quad x \text{---} \boxed{C} \text{---} y \otimes z \text{---} \boxed{D} \text{---} w := \begin{array}{c} x \text{---} \boxed{C} \text{---} y \\ z \text{---} \boxed{D} \text{---} w \end{array}$$

The units are empty wires and empty space, respectively. In this way, circuits form strict monoidal categories.

Circuit Languages as Free Monoidal Categories

When studying circuits, one fixes a gate set Σ , defined as follows.

- **Wire Types.** A set Σ_0 (*finite in practice*).
- **Primitive Gates.** A set Σ_1 (*possibly infinite, as in [4]*).
- **Gate Types.** Functions $\text{dom}, \text{cod} : \Sigma_1 \rightarrow \Sigma_0^*$.

Free Monoidal Categories. This data is often called a *monoidal signature* [10]. Signature morphisms are defined in the obvious way. It can be shown that the following adjunction exists.

$$\begin{array}{ccc} & \xrightarrow{F_M} & \\ \mathbf{MSig} & \perp & \mathbf{MCat}^{\text{strict}} \\ & \xleftarrow{U_M} & \end{array}$$

Circuit Semantics. A semantic interpretation for a gate set Σ is consequently defined by a signature morphism $\Sigma \rightarrow U_M(\mathcal{C})$. In the case of quantum circuits, $\mathcal{C}$ is often **Vect** or **CPTP**.

An Aside on Free Monoidal Categories

*It can be asked whether **MSig** admits a graph-theoretic description. It turns out that the objects in **MSig** are equivalent to 2-polygraphs with a single vertex. Such polygraphs generate 2-categories with a single object [2]. These correspond to strict monoidal categories via the delooping of monoidal categories.*

From Sets to Signatures: Circuit Presentations

Circuit Presentations. Monoid presentations can be generalized to monoidal categories. Such presentations have applications to circuit optimization [8], simulation [11, 12], and verification [9].

Relations. A relation is a pair of morphisms (u, v) in $U_M(\mathcal{C})$ such that $\text{dom}(u) = \text{dom}(v)$ and $\text{cod}(u) = \text{cod}(v)$. A set of a relations form a monoidal signature R with projections $l, r : R \rightarrow U_M(\mathcal{C})$.

Monoidal Presentations. A presentation of a monoidal category $\mathcal{C}$ is a signature Σ and a set R of relations such that $\mathcal{C} \cong \langle \Sigma | R \rangle$ as defined by the following coequalizer.

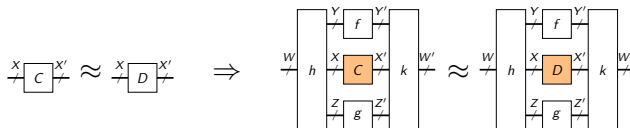
$$F_M(R) \begin{array}{c} \xrightarrow{l^*} \\ \xleftarrow{r^*} \end{array} F_M(\Sigma) \xrightarrow{\pi_E} \langle \Sigma | R \rangle$$

Circuit Presentations as Rewriting Systems

As with monoids, $\pi_E : F_M(\Sigma) \rightarrow \langle \Sigma | E \rangle$ induces an equivalence relation ($\approx$) with the following congruence properties.

- If $u \approx v$ and (f, u, g) composable, then $f \circ u \circ g \approx f \circ v \circ g$.
- If $u \approx v$, then $f \otimes u \otimes g \approx f \otimes v \otimes g$.

Then π_E can be understood in terms of circuit rewriting.



A Technical Remark. Adding relations to a monoidal presentation can be seen as gluing 3-cells to the 2-polygraph of Σ . Realizing the presentation is then taking the 2-truncation of the polygraph [2].

This is why objects remain free after computing the coequalizer [3].

Example: Why Rewriting Systems?

Assume that $C \cong \langle \Sigma | E \rangle$ and $\llbracket - \rrbracket : \Sigma \rightarrow \mathcal{D}$ induces an isomorphism.

Circuit Optimization. Given a circuit $C \in F_M(\Sigma)$, find a circuit $D \in F_M(\Sigma)$ such that $\llbracket C \rrbracket = \llbracket D \rrbracket$ with C more *efficient* than D .

Local Optimization. An optimization that replaces each instance of $g \in \Sigma_1$ with some $C \in F_M(\Sigma)$. This defines a monoidal functor $\text{opt} : F_M(\Sigma) \rightarrow F_M(\Sigma)$ via the signature $\tau : \Sigma \rightarrow U_M(F_M(\Sigma))$.

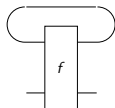
$$\tau(x) = \begin{cases} C & \text{if } x = g \\ x & \text{otherwise} \end{cases}$$

Correctness. We know that $\llbracket F(-) \rrbracket = \llbracket - \rrbracket$ if and only if $F(-)$ factors through $\mathcal{C}$. By the universal property of quotients, $F(-)$ factors through $\mathcal{C}$ if and only if $F(u) = F(v)$ for all $(u, v) \in E$.

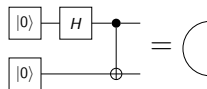
A Personal Interest: Extending Circuit Languages

Often times, simply working with monoidal circuit diagrams is insufficient. For example, wires can often (but not always) be swapped. In this case, the category is symmetric.

A question of personal interest to me, is what properties must be added to a category to capture features of certain circuit languages.



Feedback via traces [7].



Entanglement as duals [1].

$$\left[\begin{array}{c} n \\ \text{---} \end{array} \boxed{C_n} \begin{array}{c} n \\ \text{---} \end{array} \right] = \coprod_{n \in \mathbb{N}} X^{\otimes n}$$

Families via dependent types [6].

$$\llbracket \text{---} \boxed{R_Z(\theta)} \text{---} \rrbracket = \left(\theta \mapsto \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} \right)$$

Parameters via enrichment.

Conditional Operations in Circuits

A common pattern in circuit design is to condition the execution of a circuit $f : X \rightarrow Y$ on whether a *control wire* is in state $|1\rangle$.

$$\left[\begin{array}{c} \text{Control} \text{---} \bullet \text{---} \\ \text{Target} \text{---} \boxed{f} \end{array} \right] (|c\rangle \otimes |x\rangle) = \begin{cases} |1\rangle \otimes [f](|x\rangle) & \text{if } c = 1 \\ |c\rangle \otimes |x\rangle & \text{otherwise} \end{cases}$$

In quantum computation, given a unitary $U : \mathbb{C}^n \rightarrow \mathbb{C}^n$, this corresponds to the direct sum of matrices with the identity.

$$(I_n \oplus U)(|0\rangle \otimes |x\rangle) = |0\rangle \otimes |x\rangle \quad (I_n \oplus U)(|1\rangle \otimes |x\rangle) = |1\rangle \otimes U|x\rangle$$

More generally, f can be *controlled* by the state of a control wire, without modifying the state of the control wire.

$$C(f : 2 \rightarrow 2) =$$

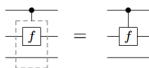
A Functorial Approach: Control Functors

A functorial description of controlled operations was provided by Delorme and Perdrix in [5].

Setting. Let $(\mathcal{C}, \otimes, \mathbb{I}, \sigma, \dots)$ be a symmetric monoidal category such that $(\text{Ob}(\mathcal{C}), \otimes, \mathbb{I}) \cong (\mathbb{N}, +, 0)$.

Definition. An ordinary functor $C : \mathcal{C}_{\text{Endo}} \rightarrow \mathcal{C}_{\text{Endo}}$ such that $C(n) = n + 1$ and the following three properties hold.

$$C(f \otimes \text{id}_1) = C(f) \otimes \text{id}_1$$



$$C(C(f)) \circ (\sigma_{1,1} \otimes \text{id}_n) = (\sigma_{1,1} \otimes \text{id}_n) \circ C(C(f))$$



$$C((\text{id}_k \otimes \sigma_{1,1} \otimes \text{id}_\ell) \circ f \circ (\text{id}_k \otimes \sigma_{1,1} \otimes \text{id}_\ell)) = (\text{id}_{k+1} \otimes \sigma_{1,1} \otimes \text{id}_\ell) \circ C(f) \circ (\text{id}_{k+1} \otimes \sigma_{1,1} \otimes \text{id}_\ell)$$



Goals of this Work-in-Progress

In [5], it was shown how to freely generate categories with control functors by viewing $C(-)$ as a constructor. This is not compatible with monoidal presentations, since such a construction freely adds new terms with new relations.

$$f \in \Sigma_1 \quad \Longrightarrow \quad \begin{array}{c} \bullet \\ | \\ \boxed{f} \end{array}, \begin{array}{c} \bullet \\ | \\ \bullet \\ | \\ \boxed{f} \end{array}, \begin{array}{c} \bullet \\ | \\ \bullet \\ | \\ \bullet \\ | \\ \boxed{f} \end{array}, \dots \in \text{Mor}(F(\Sigma))$$

I am currently working to answer the following questions.

- Are control functors determined by monoidal generators?
- When can control functors be freely defined with respect to a monoidal presentation?
- Can these answers be generalized to a wider class of functors with weaker axioms?

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Overview of this Section

Motivating Question. If F and G are control functors on $\mathcal{C}$ and $\mathcal{C}_{\text{Endo}}$ is monoidally generated by Σ , does $\forall f \in \Sigma_1, F(f) = G(f)$ imply that $F = G$?

Remark. This does not follow from the free-forget adjunction, since F and G are not monoidal functors.

Approach. The section proceeds as follows.

- Monoidal generating sets are formalized for **BrCat**^{strict}.
- An inclusion **MSig** $\rightarrow$ **Digraph** is constructed.
- The evaluation of an ordinary functor on a monoidal generating set is formalized via this inclusion.
- The notion of control functor is weakened, and the question is answered in the positive for this wider class of functors.

Defining Monoidal Generators

Begin by observing the following sequence of adjunctions.

$$\mathbf{MSig} \begin{array}{c} \xrightarrow{F_M} \\ \perp \\ \xleftarrow{U_M} \end{array} \mathbf{MCat}^{\text{strict}} \begin{array}{c} \xrightarrow{F_{Br}} \\ \perp \\ \xleftarrow{U_{Br}} \end{array} \mathbf{BrCat}^{\text{strict}}$$

Given a braided monoidal category $\mathcal{C}$, we can find its underlying monoidal signature $U_M(U_{Br}(\mathcal{C}))$.

For each subobject $i : \Gamma \rightarrow U_M(U_{Br}(\mathcal{C}))$, there exists a unique morphism $j : F_M(\Gamma) \rightarrow U_M(U_{Br}(\mathcal{C}))$ such that the following commutes.

$$\begin{array}{ccc} F_M(\Gamma) & & \\ \eta_\Gamma \uparrow & \searrow U_M(j) & \\ \Gamma & \xrightarrow{i} & U_M(U_{Br}(\mathcal{C})) \end{array}$$

We say that Γ *monoidally generates* $\mathcal{C}$ if j is full and **surjective**.

Example: Generating the Unitary Matrices

Let $\psi(\theta) = e^{i\theta}$ and consider the following matrices.

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad R_Z(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} \quad CX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

It can be shown that the following set of matrices monoidally generate the category of unitary matrices with rank a power of 2.

$$\Sigma_1 = \{\psi(\theta), R_Z(\theta) : \theta \in [-\pi, \pi)\} \cup \{H, CX\}$$

Notice that the symmetries must also be generated by Σ . Since this is a prop category, it suffices to construct the generating permutation matrix $\beta : \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^2 \otimes \mathbb{C}^2$.

$$\beta = CX \circ (H \otimes H) \circ CX \circ (H \otimes H) \circ CX$$

Interpreting Monoidal Signatures as Digraphs

Idea. Assume that Σ is a monoidal signature. $\text{dom} : \Sigma_1 \rightarrow \Sigma_0^*$ and $\text{cod} : \Sigma_1 \rightarrow \Sigma_0^*$, then Σ can be viewed as a graph with vertices Σ_0^* and edges Σ_1 . Then $\gamma(\Sigma) := (\Sigma_0^*, \Sigma_1^*, \text{dom}, \text{cod})$ is a digraph.

Remark. A signature morphism from Σ to Γ is definitionally a digraph morphism from $\gamma(\Sigma)$ to $\gamma(\Gamma)$. Then the assignment $\gamma(-)$ defines a functor $\gamma : \mathbf{MSig} \rightarrow \mathbf{Digraph}$.

Interpretation. The category $F_C(\gamma(\Sigma))$ is the non-monoidal wide subcategory of $F_M(\Sigma)$ generated by the morphisms in Σ under only sequential composition. If $f, g \in \Sigma_1$ with $\text{dom}(g) = \text{cod}(f)$, then...

$$\boxed{f} \text{---} \boxed{g} \text{---} \in F_C(\gamma(\Sigma)) \quad \begin{array}{c} \boxed{f} \\ \boxed{g} \end{array} \notin F_C(\gamma(\Sigma)) \quad \underline{\boxed{f}} \notin F_C(\gamma(\Sigma))$$

Evaluating Ordinary Functors on Monoidal Generators

The following diagram can then be constructed.

$$\begin{array}{ccc}
 \mathbf{Cat} & \xleftarrow{U} & \mathbf{MCat}^{\text{strict}} \\
 F_C \uparrow & \swarrow \text{ev} & \uparrow F_M \\
 \mathbf{Digraph} & \xleftarrow{\gamma} & \mathbf{MSig}
 \end{array}$$

A Second Digraph. Let $\Sigma \in \mathbf{MSig}$ and $F_M(\Sigma) = (\mathcal{C}, \otimes, \mathbb{I})$. It is not hard to verify $U(F_M(\Sigma)) = \mathcal{C}$ and $U_C(\mathcal{C}) = (\Sigma_0^*, \text{Mor}(\mathcal{C}))$.

An Embedding. By the adjunction $F_M \dashv U_M$, there exists a map $\eta_\Sigma : \Sigma \hookrightarrow U_M(F_M(\Sigma)) = (\Sigma_0^*, \text{Mor}(\mathcal{C}))$. Define the embedding $i : \gamma(\Sigma) \hookrightarrow U_C(\mathcal{C})$ such that i_0 is the identity and $i_1 = (\eta_\Sigma)_1$.

Evaluating. Let $\text{ev}_\Sigma : F_C(\Gamma) \rightarrow \mathcal{C}$ by the unique morphism defined by i . This universal property can be used to prove the following.

$$\forall f : \Sigma \rightarrow \Gamma, \text{ev}_\Gamma \circ F_C(\gamma(f)) = U(F_M(f)) \circ \text{ev}_\Gamma$$

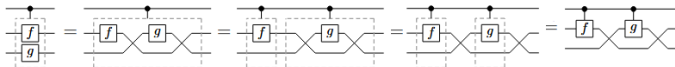
A Formula for Evaluating Control Functors

Assume that $C : \mathcal{C}_{\text{Endo}} \rightarrow \mathcal{C}_{\text{Endo}}$ is a control functor and Σ is a monoidal generating set for $\mathcal{C}$.

Problem. Define $g_C = C(i(g))$ for each $g \in \Sigma_1$. If $f \in \text{Mor}(\mathcal{C})$, then can $C(f)$ be expressed purely in terms of $\{g_C : g \in \Sigma_1\}$?

Solution. The problem can be solved by an inductive formula.

- ① **Decomposition.** Since Σ generates $\mathcal{C}$, then there exists a word $w \in F_M(\Sigma)$ such that $i(w) = f$.
- ② **Base Cases.** There are two primitive terms.
 - If $w = 1_n$, then $C(w) = 1_{n+1}$.
 - If $w \in \Sigma_1$, then $C(w) = w_C$.
- ③ **Inductive Cases.** Sequential composition is determined by functoriality. Parallel composition is derived in [5].



Weakening the Notion of Control Functors

The previous calculation shows how to evaluate a control functor on an arbitrary morphism in $\mathcal{C}$ using only its evaluation on the monoidal generators for $\mathcal{C}$.

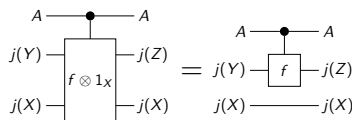
However, this proof followed from a weaker set of axioms than those used to define a control functor. The following properties were not used in the proof at all.

- $\mathcal{C}$ is a PROP category.
- Iterated application of $\mathcal{C}$ is symmetric about the control wires.
- $\mathcal{C}$ is endomorphic and only applies to endomorphisms.
- $\mathcal{C}$ respects conjugation by **arbitrary** symmetries.

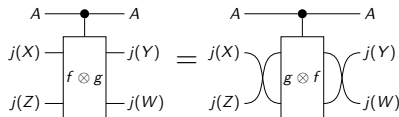
Introducing Precontrol Functors

Let $(\mathcal{C}, \otimes, \mathbb{I}, \beta)$ and $(\mathcal{D}, \boxtimes, \mathbb{I}, \sigma)$ be strict braided monoidal cats. A *precontrol functor* is an ordinary functor $F : \mathcal{C} \rightarrow \mathcal{D}$ satisfying the following properties.

- 1 There exists an object $A \in \mathcal{D}$ and a monoid morphism $j : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$ such that $F(X) = A \otimes j(X)$.
- 2 For each $X \in \text{Ob}(\mathcal{C})$ and $f \in \mathcal{C}(Y, Z)$, the following holds.



- 3 For each $f \in \mathcal{C}(X, Y)$ and $g \in \mathcal{C}(Z, W)$, the following holds.



Monoidal Generators Determine Precontrol Functors

Now we can put everything together...

- ① Let $\mathcal{C}$, be a braided monoidal category with a collection of monoidal generators Σ and inclusion $j : F(\Sigma) \rightarrow U_B(\mathcal{C})$.
- ② Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{C} \rightarrow \mathcal{D}$ be precontrol functors.

Informally, we would like to ask that $F(f) = G(f)$ for all $f \in \Sigma_1$. We can make this precise by the following equation.

$$F \circ U(j) \circ \text{ev}_\Sigma = G \circ U(j) \circ \text{ev}_\Sigma$$

As in the case of control functors, the evaluation of F and G on the generators in Σ_1 will inductively determine the evaluation of F and G on all morphisms in $\mathcal{C}$. In other words, the following is true.

$$(F \circ U(j) \circ \text{ev}_\Sigma = G \circ U(j) \circ \text{ev}_\Sigma) \quad \Rightarrow \quad (F = G)$$

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Free Generation: Identifying the Requirements

Let Σ monoidally generate $\mathcal{C}$ and $\tau : \gamma(\Sigma) \rightarrow U_{\mathcal{C}}(U(U_B(\mathcal{D})))$.

Before τ could possibly define a control functor, one must check that there exists an $A \in \text{Ob}(\mathcal{D})$ and $j : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$ such that $\tau(X) = A \otimes j(X)$ for each $X \in \Sigma_0^*$.

Problem. The last section showed how to evaluate a precontrol functor $F(-)$ in terms of γ . However, the evaluation of $F(f)$ depended on the choice of a $w \in F(\Sigma)$ such that $f = i(w)$.

Solution. To yield a well-defined precontrol functor, $F(f)$ must not depend on the choice of w . There are two sorts of dependencies.

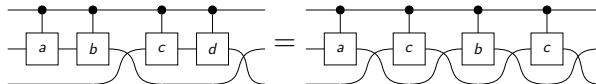
- **Syntactic.** Bifunctoriality, associativity, and unitality.
- **Semantic.** Relations between generators in Σ .

Remark. The category $F_M(\Sigma)$ is not braided.

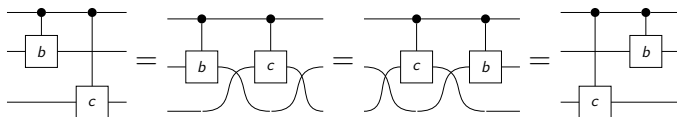
Obstructions: Respecting Bifunctionality

Structural Relation. $(b \circ a) \otimes (d \circ c) = (b \otimes d) \circ (a \otimes c)$.

Precontrol Relation. Assume $f = i(w)$ and $w = (b \circ a) \otimes (d \circ c)$. Then the following equation must hold.



Notice that the following relation suffices to respect bifunctionality.

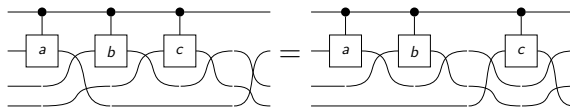


Moreover, this equation follows from bifunctionality, in the case where (a, d) are both identities.

Obstructions: Respecting Associativity

Structural Relation. $a \otimes (b \otimes c) = (a \otimes b) \otimes c$.

Precontrol Relation. Assume $f = i(w)$ and $w = a \otimes (b \otimes c)$. Then the following equation must hold.

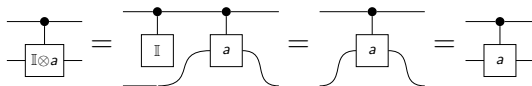


It is not hard to see that the left-hand side can be transformed into the right-hand side via untangling of the wires.

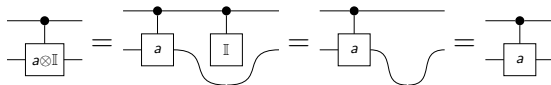
Obstructions: Respecting Unitality

Structural Relation. $\mathbb{I} \otimes a = a = a \otimes \mathbb{I}$.

Precontrol Relation. Assume $f = i(w)$ and $w = 1 \otimes a$. Then the following equations holds.



Likewise, if $w = a \otimes 1$, then the following equation holds.



Then this construction respects associativity by definition.

The Free Precontrol Functor

It remains to be shown that the f decomposition respects the relations in $\mathcal{C}$. To make this problem tractable, we will assume that $\mathcal{C} \cong \langle \Sigma | E \rangle$. Then it suffices to check the following condition.

$$u \approx_E v \quad \Rightarrow \quad \begin{array}{c} \bullet \\ | \\ \boxed{u} \end{array} = \begin{array}{c} \bullet \\ | \\ \boxed{v} \end{array}$$

It can then be shown by well-founded induction that the tensor decompositions commute if and only if the property holds for the monoidal generators.

$$\forall (f, g) \in \Sigma_1 \times \Sigma_1, \quad \begin{array}{c} \bullet \quad \bullet \\ | \quad | \\ \boxed{f} \quad \boxed{g} \end{array} = \begin{array}{c} \bullet \quad \bullet \\ | \quad | \\ \boxed{g} \quad \boxed{f} \end{array}$$

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Control Symmetry is Compositional

Assume that $F : \mathcal{C}_{\mathbf{Endo}} \rightarrow \mathcal{C}_{\mathbf{Endo}}$ is a precontrol functor. In order for F to be a control functor, it must satisfy control symmetry.

$$\forall f \in \text{Mor}(\mathcal{C}), \quad \begin{array}{c} \bullet \\ \bullet \\ \text{---} f \text{---} \end{array} = \begin{array}{c} \text{---} \bullet \\ \text{---} \bullet \\ \text{---} f \text{---} \end{array}$$

However, control symmetry is a compositional property.

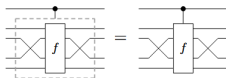
$$\begin{array}{c} \bullet \\ \bullet \\ \text{---} g \circ f \text{---} \end{array} = \begin{array}{c} \bullet \quad \bullet \\ \bullet \quad \bullet \\ \text{---} f \text{---} \quad \text{---} g \text{---} \end{array} = \begin{array}{c} \bullet \quad \bullet \\ \bullet \quad \bullet \\ \text{---} f \text{---} \quad \text{---} g \text{---} \end{array} = \begin{array}{c} \bullet \quad \bullet \\ \bullet \quad \bullet \\ \text{---} f \text{---} \quad \text{---} g \text{---} \end{array} = \begin{array}{c} \bullet \quad \bullet \\ \bullet \quad \bullet \\ \text{---} f \text{---} \quad \text{---} g \text{---} \end{array} = \begin{array}{c} \bullet \\ \bullet \\ \text{---} g \circ f \text{---} \end{array}$$

This means that it suffices to check that each generator satisfies control symmetry, where $\mathcal{C}_{\mathbf{Endo}} \cong \langle \Sigma | E \rangle$.

$$\forall g \in \Sigma_1, \quad \begin{array}{c} \bullet \\ \bullet \\ \text{---} g \text{---} \end{array} \approx_E \begin{array}{c} \text{---} \bullet \\ \text{---} \bullet \\ \text{---} g \text{---} \end{array}$$

Target Symmetry is Compositional

The functor $F : \mathcal{C}_{\mathbf{Endo}} \rightarrow \mathcal{C}_{\mathbf{Endo}}$ must also satisfy target symmetry.



Since every monoidal category is a premonoidal category (see [?]), then it is possible to decompose every word in $F_M(\Sigma)$ as the sequential composition of zero or more terms of the following form.

$$1_X \otimes f \otimes 1_Y \text{ where } (X, f, Y) \in \Sigma_0^* \times \Sigma_1 \times \Sigma_0^*$$

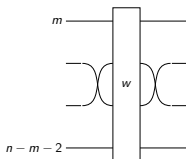
Similar to control symmetry, target symmetry is compositional. Then it suffices to prove target symmetry for each term of the form $1_n \otimes f \otimes 1_m$, where $(n, f, m) \in \mathbb{N} \otimes \Sigma_1 \otimes \mathbb{N}$.

Remark. This family of relations is infinite, even if Σ is finite.

Target Symmetry Permits Finite Verification

Given a finite presentation $\langle \Sigma | E \rangle$ for $\mathcal{C}_{\mathbf{Endo}}$, it is possible to reduce target symmetry in a finite number of rewriting problems. This reduction holds even if $(\text{Ob}(\mathcal{C}), \otimes, \mathbb{I}) \not\cong \mathbb{N}$.

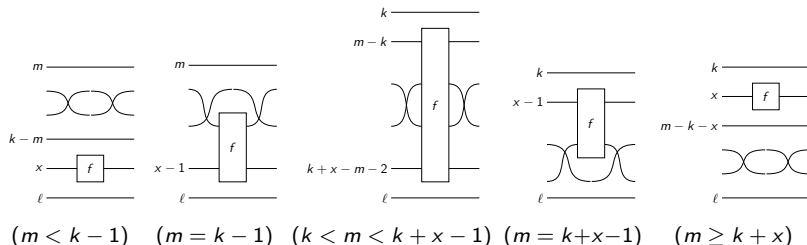
Combinatorial Enumeration. Assume that $w \in \mathcal{C}(n, n)$ is of the form $1_k \otimes f \otimes 1_\ell$. Consider the following circuit for some $m \in \mathbb{N}$.



This family of circuits can be classified based on how the braiding intersects with f . It can then be shown that many of the cases already follow from the axioms of a precontrol functor.

Classifying Premonoidal Generating Terms

Let $m \in \mathbb{N}$ as on the previous slide and $f \in \mathcal{C}(x, x)$ so that $n = k + x + \ell$. There are several cases to consider.

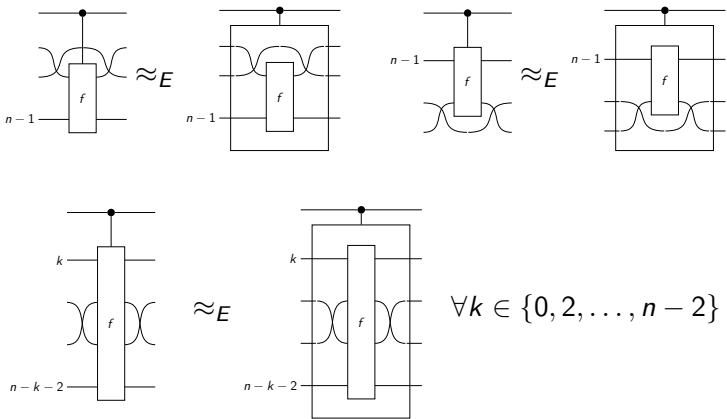


When the symmetries do not intersect with f (or $x = 1$), it is possible to prove target symmetry from the precontrol axioms.

Conclusion. It suffices to check target symmetry for the $x + 1$ possible ways the braiding intersects with each $f \in \Sigma_1$ (for $x \geq 2$).

Illustrating the Three Cases

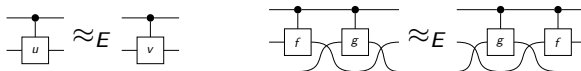
Assume that $f \in \Sigma_1$ and $n = \text{dom}(f) \geq 2$. The three cases described on the preceding slide are illustrated below.



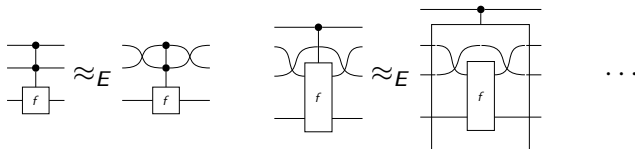
- 1 Background, Motivation, and Goals
- 2 Monoidal Generators and Generalized Control Functors
- 3 Free Precontrol Functors from Monoidal Generators
- 4 From Precontrol Functors to Control Functors
- 5 Conclusion

Summary of Results

- Precontrol functors are determined by monoidal generators.
- If $\mathcal{C} \cong \langle \Sigma | E \rangle$, then a signature $\tau : \gamma(\Sigma) \rightarrow F_C(U(F_M(\Sigma)))$ defines a precontrol functor if-and-only-if $\tau_0(-) = A \otimes j(-)$ and the following hold for all $(u, v) \in E$ and $(f, g) \in \Sigma_1 \times \Sigma_1$.



- This precontrol functor is a control functor if and only if the following hold for all $f \in \Sigma_1$ and $g \in \Sigma_1$ with $\text{dom}(g) > 1$.



Future Work

- Formalize the adjunction for precontrol functors.
- Use this adjunction to construct non-obvious examples of control functors.
- Extend the precontrol characterization of control functors to control functors with adjectives (e.g., polycontrol functors and pointed control functors).
- Explore *good* definitions of control functors.
- Reasoning about mixed classical-quantum control.
- Interpreting the controlled SWAP gate.

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